

Muttavarapu Pavan Aditya

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Education

Manipal Institute of Technology

B.Tech. in Computer Science and Engineering (AI/ML); CGPA: 8.66/10.0

Bengaluru, India

Expected 2027

Experience

Handshake AI — Project Dynamo

Software Engineer (Contract)

Remote

Jul 2026 – Present

- Designed benchmark tasks that measure whether AI coding agents can isolate root causes, planting interacting defects across data processing, numerical computation, scheduling, and dependency handling; frontier agents averaged **0.2/1.0**.
- The agents often produced fixes that looked correct but failed on edge cases, so wrote **pytest** checks against reference implementations, including numerical checks down to **1e-5**.
- A dependency graph with **5 million events** became a CI bottleneck, so reworked its generation while preserving topological order and preventing cycles.

SwarmLens

AI & DevOps Engineer Intern

Kochi, India (Remote)

Feb 2026 – May 2026

- Built a meeting-processing application in Python and WhisperX that turned recorded calls into speaker-attributed transcripts, action items, and decisions.
- The first pipeline handled extraction and verification together, so split it into **4 specialized components**, reducing the measured hallucination rate by approximately **55%**.
- As the application moved behind a frontend, configured NGINX for routing and reduced average API response time from **420 ms to 300 ms**; added KPI dashboards to monitor regressions.

Projects

Streaming Platform Backend | Node.js, Kafka, Redis, PostgreSQL, Microservices

[/streamingbackend](#)

- Started with a monolithic streaming backend, then split it into **6 services** when playback, watch-history, notifications, subscriptions, and related workloads became difficult to scale independently.
- The services became tightly coupled through synchronous calls, so introduced Kafka for asynchronous event processing and Saga orchestration for failures across subscription and billing workflows.
- Read-heavy endpoints remained slow after the split, so cached high-frequency paths in Redis and reduced response time from **1.2 s to under 150 ms**; also built a recommendation service supporting **20+ natural-language search intents** with responses under **2 s**.

AI Talent Intelligence & Live Interview Platform | Python, FastAPI, Oracle 26ai, WebRTC

[/talentintelligence](#)

- Defined data models for candidates, jobs, applications, and ATS reports, then built resume-processing pipelines that extracted **10+ attributes** into Oracle 26ai; added Text-to-SQL support for **20+ query patterns**.
- Recommendations varied between runs, so built a benchmarking harness to compare outputs and used the failures to refine the pipeline, improving run-to-run consistency by approximately **25%**.
- Built a real-time interview workspace with Monaco Editor and WebRTC for collaborative coding, video, and screen sharing, with approximately **1.8 s** end-to-end voice latency; automated post-interview reports in under **3 s**, reducing manual feedback effort by approximately **70%**.

Enterprise Knowledge Retrieval Platform | Python, FastAPI, Qdrant, RAGAS

[/documentsearch](#)

- Started with the problem of making enterprise documents searchable across different file types, then built an ingestion pipeline supporting **5 formats**, paragraph-based chunking, Jina embeddings, and Qdrant vector search.
- Retrieval quality could not be judged reliably from individual responses, so built a **21-sample** evaluation pipeline using RAGAS, achieving **95% Faithfulness** while measuring Answer Relevancy, Context Precision, Context Recall, and Answer Correctness.
- Added semantic reranking and history-aware query planning when basic retrieval was insufficient, and built a Streamlit dashboard that captured contexts, tool calls, and responses to trace failures and compare evaluation results.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash

Backend & Distributed Systems: FastAPI, Node.js, Express.js, REST, WebSockets, WebRTC, Kafka, Redis, Microservices, Saga

AI/ML: LangChain, RAG, RAGAS, Qdrant, WhisperX, Text-to-SQL, OpenAI and Gemini APIs

Infrastructure & Tools: Docker, NGINX, Git, pytest, CI/CD

Databases: PostgreSQL, MySQL, MongoDB, Neo4j, Oracle 26ai

Achievements & Certifications

Solved **600+ LeetCode problems** and ranked among the **top 10 students college-wide** in DSA on GeeksforGeeks.

Certifications: Interview Preparation (Meta) | Advanced Backend (Namaste Dev) | Data Structures & Algorithms (takeUforward, Raj Vikramaditya) | Introduction to Programming with Python and Java (University of Pennsylvania)